

A Major Project Report on

Deep Learning-Based Sleep Disorder Detection Using EEG Signal Analysis

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CERTIFICATE

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ABSTRACT

Sleep disorders significantly impact human health and quality of life. Traditional diagnosis methods such as polysomnography (PSG) are expensive, time-consuming, and require expert intervention. This project presents an automated deep learning-based framework for detecting sleep disorders using EEG signals. The proposed system utilizes a hybrid architecture combining Convolutional Neural Networks (CNN), Bidirectional Long Short-Term Memory (BiLSTM), and Multi-Head Self-Attention mechanisms to extract spatial, temporal, and contextual features from EEG signals. The system processes EEG data through preprocessing techniques such as filtering, resampling, and segmentation, followed by extraction of multi-domain features including spectral, statistical, and wavelet-based features. The model is trained on EEG recordings categorized into four classes: Normal, Insomnia, Narcolepsy, and Sleep Disordered Breathing. To address class imbalance, techniques such as SMOTE and class-weighted loss are employed. The proposed model achieves an accuracy of 92.24% and demonstrates strong generalization capabilities. The results indicate that deep learning-based approaches can effectively automate sleep disorder diagnosis and have potential for real-world clinical applications.

Keywords: *Sleep disorder detection, EEG signals, Deep learning, Convolutional Neural Network, BiLSTM, Multi-Head Self-Attention, Hybrid architecture, Class imbalance, Automated diagnosis*

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SYMBOLS & ABBREVIATIONS

S.No	Symbol / Abbreviation	Description
1	EEG	Electroencephalography
2	CNN	Convolutional Neural Network
3	BiLSTM	Bidirectional Long Short-Term Memory
4	PSG	Polysomnography
5	SVM	Support Vector Machine
6	SMOTE	Synthetic Minority Over-sampling Technique
7	MHA	Multi-Head Attention
8	FFT	Fast Fourier Transform
9	DWT	Discrete Wavelet Transform
10	PSD	Power Spectral Density
11	RNN	Recurrent Neural Network
12	GRU	Gated Recurrent Unit
13	SDB	Sleep Disordered Breathing
14	EDF	European Data Format
15	TP	True Positives
16	TN	True Negatives
17	FP	False Positives
18	FN	False Negatives
19	Accuracy	Measure of overall correctness
20	F1-Score	Harmonic mean of precision and recall

CHAPTER 1 INTRODUCTION

Sleep is a fundamental biological process that plays a critical role in maintaining human health, cognitive function, and overall well-being. During sleep, the body undergoes essential restorative processes including tissue repair, immune system strengthening, and memory consolidation. Despite its importance, sleep disorders have become increasingly prevalent in modern society, affecting a significant proportion of the global population across all age groups. Conditions such as insomnia, narcolepsy, and sleep-disordered breathing not only reduce the quality of life but also contribute to a wide range of serious health complications including cardiovascular diseases, metabolic disorders, obesity, and cognitive impairments.

The clinical assessment of sleep disorders has traditionally relied on polysomnography (PSG), which is considered the gold standard for sleep analysis. PSG involves the simultaneous recording of multiple physiological signals, including electroencephalography (EEG), electrooculography (EOG), electromyography (EMG), respiratory signals, and oxygen saturation, over an entire night in a clinical laboratory setting. While PSG provides comprehensive information about sleep architecture and physiological parameters, it is a highly resource-intensive method that requires specialized equipment, trained technicians, and overnight laboratory visits. The cost and complexity associated with PSG make it impractical for large-scale screening, long-term monitoring, or deployment in resource-constrained environments.

Among the various physiological signals recorded during sleep, the electroencephalogram (EEG) has emerged as the most informative modality for characterizing sleep stages and identifying pathological patterns associated with sleep disorders. EEG measures the electrical activity of the brain through electrodes placed on the scalp, capturing oscillatory patterns that reflect different states of neurological activity. During normal sleep, the EEG signal transitions through distinct stages characterized by specific frequency components: slow-wave oscillations associated with deep non-rapid eye movement (NREM) sleep, sleep spindles and K-complexes during stage N2, and desynchronized high-frequency activity during rapid eye

movement (REM) sleep. Deviations from these normal patterns serve as reliable biomarkers for identifying various sleep disorders.

The increasing availability of large-scale EEG datasets, combined with significant advances in computational hardware and deep learning algorithms, has created new opportunities for automated sleep disorder detection. Machine learning and deep learning approaches offer the potential to analyze complex EEG patterns objectively and efficiently, without the need for expert clinical supervision. These automated systems can reduce diagnostic delays, improve consistency of interpretation, and extend access to diagnostic capabilities in settings where clinical experts are unavailable.

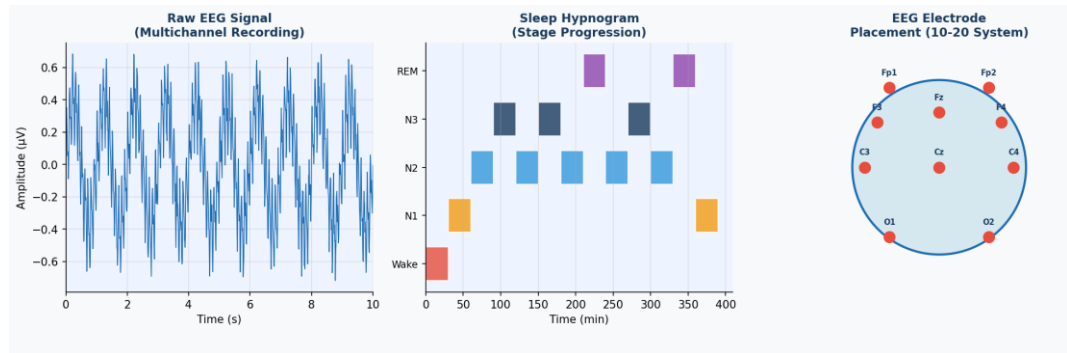


Fig 1.1: EEG Signal Recording and Sleep Stage Overview

Figure 1.1 provides a comprehensive overview of the EEG recording process and its relationship to sleep stage classification. The figure illustrates how multichannel EEG signals are recorded from standardized electrode placements on the scalp, capturing the oscillatory brain activity patterns that correspond to different sleep stages. The characteristic waveform morphologies associated with each sleep stage are depicted, highlighting the transitions between wakefulness, light sleep, deep sleep, and REM sleep.

The complexity of EEG signals presents significant challenges for automated analysis. Unlike other physiological signals, EEG is characterized by high dimensionality, non-stationarity, subject-specific variability, and susceptibility to noise and artifacts from ocular movements, muscle activity, and electrical interference. The

overlap in EEG characteristics between adjacent sleep stages and between normal and pathological patterns further complicates automated classification. Additionally, the scarcity of labeled EEG data from clinical populations and the inherent class imbalance in sleep disorder datasets pose challenges for supervised learning approaches.

Early computational approaches to sleep disorder detection primarily relied on handcrafted feature extraction combined with classical machine learning algorithms. These methods involved domain experts manually selecting relevant features from the time domain, frequency domain, and time-frequency domain of EEG signals, which were then used as inputs to classifiers such as Support Vector Machines, Random Forests, and k-Nearest Neighbors. While these approaches demonstrated initial promise, they were inherently limited by the quality and completeness of the manually designed features. The high dimensionality and complexity of EEG signals make it difficult to identify a compact set of features that generalizes well across different individuals and recording conditions.

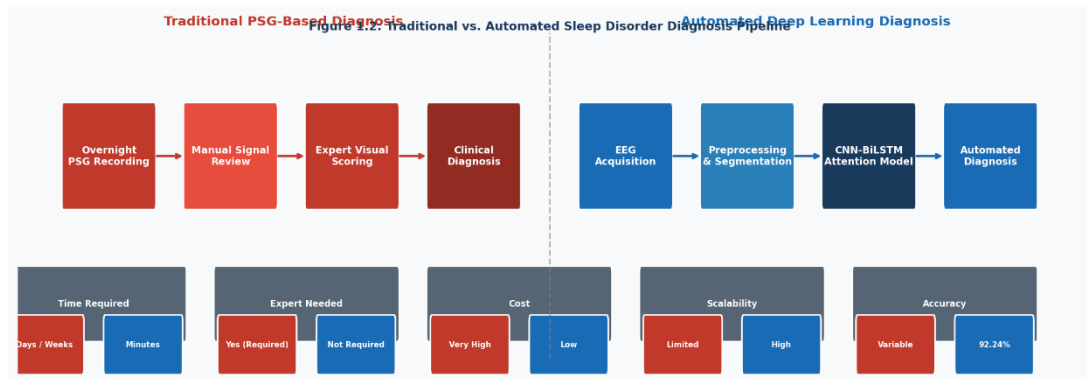


Fig 1.2: Traditional vs. Automated Sleep Disorder Diagnosis Pipeline

Figure 1.2 illustrates the contrast between traditional clinical diagnostic workflows and automated computational approaches for sleep disorder detection. The traditional pipeline involves overnight PSG recording, manual visual scoring of sleep stages by trained clinicians, and subsequent clinical interpretation. This process is time-consuming, subjective, and dependent on clinician expertise. In contrast, the automated pipeline processes raw EEG signals through computational analysis stages including

preprocessing, feature extraction, and machine learning classification to produce diagnostic outputs rapidly and objectively.

The introduction of deep learning techniques marked a significant advancement in automated EEG analysis by enabling models to learn hierarchical feature representations directly from raw or minimally preprocessed data. Convolutional Neural Networks (CNNs) were among the first deep learning architectures applied to EEG-based sleep analysis, leveraging their ability to capture local spatial patterns and temporal correlations within EEG sequences. Models such as DeepSleepNet demonstrated that CNNs could achieve performance comparable to expert scorers in sleep stage classification tasks. However, CNNs operate on fixed-length input windows and may not fully capture long-range temporal dependencies that span multiple EEG epochs.

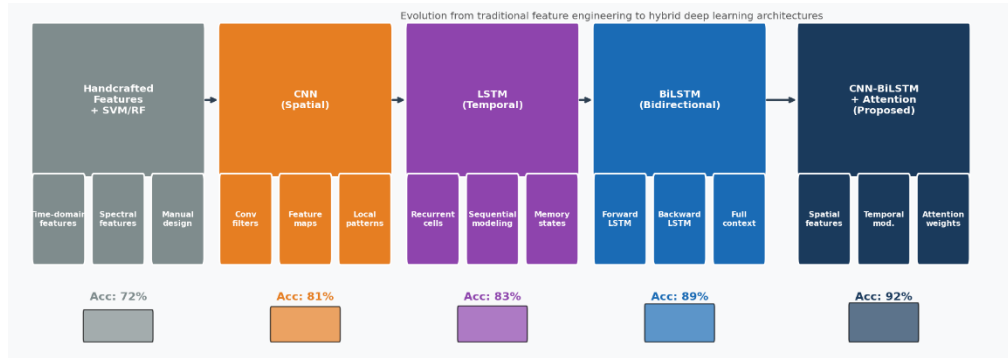


Fig 1.3: Deep Learning Approaches for EEG-Based Sleep Analysis

Recurrent Neural Networks (RNNs) and their advanced variants, including Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRUs), were subsequently applied to model the sequential dependencies in EEG sleep staging. These models process EEG sequences step-by-step, maintaining a memory state that enables the modeling of temporal dynamics across sleep stage transitions. Bidirectional LSTM (BiLSTM) architectures further enhance this capability by processing sequences in both forward and backward directions, providing the model with contextual information from past and future time steps simultaneously.

Despite their respective strengths, standalone CNN and RNN-based models each have fundamental limitations when applied to EEG sleep analysis. CNNs excel at spatial feature extraction but lack inherent mechanisms for modeling long-range temporal dependencies. RNNs capture sequential dynamics but may not effectively extract local spatial patterns from multichannel EEG signals. Research has demonstrated that integrating these complementary approaches within a unified hybrid architecture leads to significantly improved performance in EEG classification tasks.

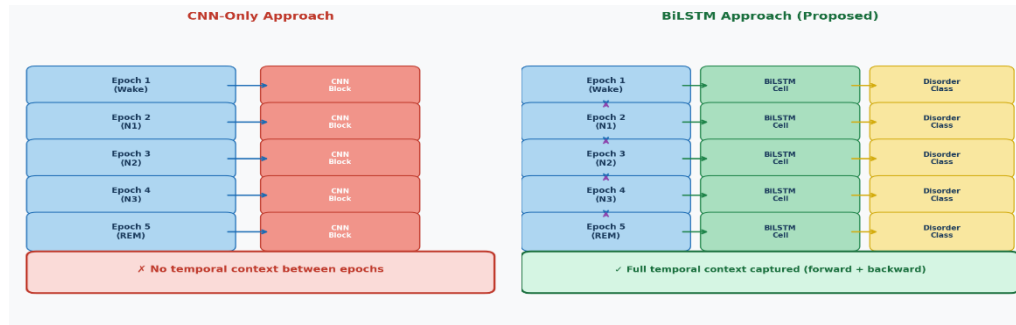


Figure 1.4: Limitations of Sequential Models for EEG Analysis

The limitations of both traditional machine learning and standalone deep learning approaches have motivated the development of hybrid architectures that combine multiple learning components within a single framework. Hybrid CNN-RNN models have shown particular promise for EEG-based sleep analysis, as they leverage the spatial feature extraction capabilities of CNNs together with the temporal modeling capabilities of recurrent networks. This combination allows the model to learn representations that capture both local signal morphology and long-range temporal dependencies, which are both essential for accurate sleep disorder classification.

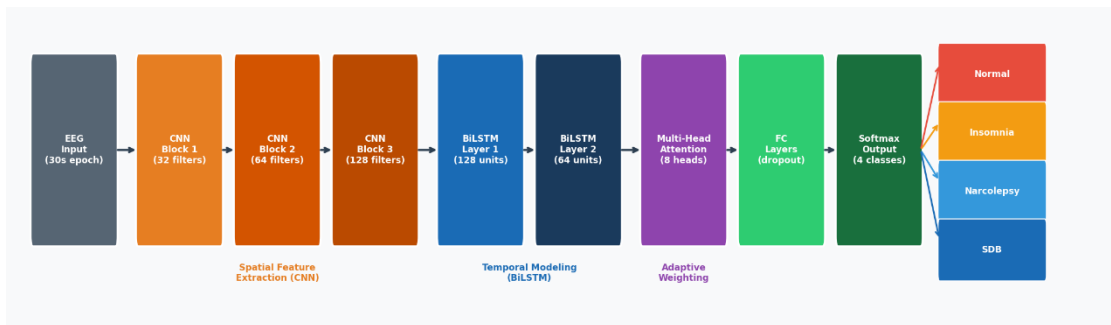


Fig 1.5: Hybrid CNN-BiLSTM Architecture Overview

A critical advancement in deep learning for sequence modeling has been the development of attention mechanisms, which enable models to selectively focus on the most relevant parts of the input when making predictions. The introduction of the Transformer architecture and Multi-Head Self-Attention mechanisms has revolutionized sequence modeling across multiple domains. In the context of EEG analysis, attention mechanisms offer the ability to identify the most diagnostically relevant temporal segments and EEG channels, enhancing both classification performance and model interpretability.



Fig 1.6: Multi-Head Self-Attention Mechanism in EEG Feature Extraction

A fundamental challenge that persists across existing approaches to automated sleep disorder detection is the limited availability of labeled EEG data from clinical populations. Obtaining high-quality labeled EEG recordings requires collaboration with sleep clinics, expert clinical annotation, and adherence to ethical research protocols, all of which limit the scale of available training data. Furthermore, clinical EEG datasets typically exhibit significant class imbalance, where common sleep stages such as NREM stage N2 are heavily overrepresented relative to stages associated with specific disorders. This imbalance can lead to biased model predictions and reduced sensitivity for detecting minority disorder categories.

These challenges highlight the need for advanced approaches that can effectively combine multiple complementary modeling strategies, leverage limited labeled data efficiently, and address the practical constraints of clinical deployment. The motivation for the present work is rooted in the need to develop a robust, accurate, and interpretable

automated system for sleep disorder detection that addresses these limitations through a carefully designed hybrid deep learning framework.

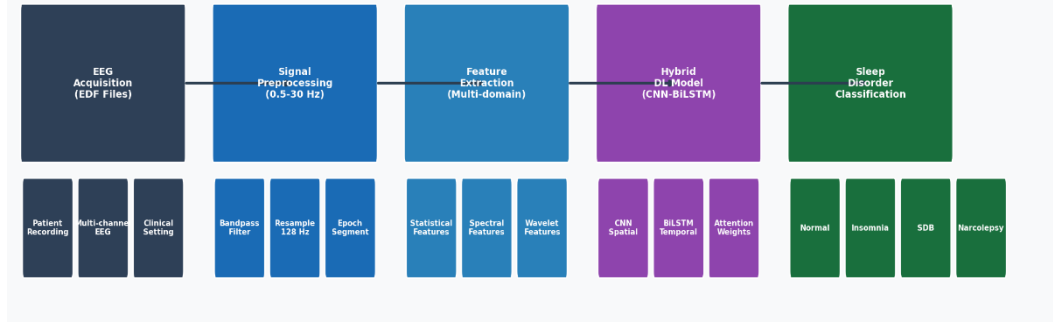


Fig 1.7: End-to-End Sleep Disorder Detection System Overview

Based on the challenges and limitations identified, this work proposes a hybrid deep learning framework for automated sleep disorder detection using EEG signals. The proposed system integrates three complementary learning components: Convolutional Neural Networks for spatial feature extraction, Bidirectional Long Short-Term Memory networks for temporal sequence modeling, and Multi-Head Self-Attention mechanisms for adaptive feature weighting and interpretability. This integrated architecture is designed to capture the full complexity of EEG sleep signals while maintaining practical efficiency suitable for clinical applications.

The primary contributions of this work can be summarized as follows. First, a comprehensive analysis of sleep disorder detection challenges and limitations of existing approaches is presented, establishing the motivation for the proposed framework. Second, a hybrid CNN-BiLSTM-Attention architecture is proposed that effectively integrates spatial, temporal, and contextual feature learning within a unified end-to-end trainable model. Third, a complete EEG signal preprocessing and feature extraction pipeline is designed to prepare raw clinical EEG data for deep learning analysis. Fourth, strategies for addressing class imbalance and improving model generalization are incorporated into the training framework. Fifth, extensive experimental evaluation is conducted on the CAP Sleep dataset to validate the effectiveness of the proposed approach compared to existing methods. These

contributions collectively advance the state of the art in automated sleep disorder detection, providing a robust and scalable solution for real-world clinical applications.

CHAPTER 2 LITERATURE SURVEY

2.1 EXISTING SYSTEM

Sleep disorder detection using computational methods has a rich history spanning several decades, evolving from simple threshold-based approaches to sophisticated deep learning architectures. The progression of methodological approaches reflects both the increasing availability of clinical EEG data and the rapid advancement of machine learning and signal processing techniques. A comprehensive review of existing systems reveals the trajectory of research progress and the persistent challenges that motivate the development of more advanced frameworks.

Early computational approaches for automated sleep stage scoring were primarily based on rule-based systems and simple pattern recognition algorithms. These systems attempted to codify the visual inspection criteria developed by sleep medicine experts, such as the Rechtschaffen and Kales (R&K) scoring manual, into algorithmic rules. While these approaches provided consistent outputs based on predefined criteria, they were inherently limited by the rigidity of rule-based reasoning and could not adapt to the variability observed in clinical EEG recordings.

The introduction of machine learning techniques represented a significant advance in automated sleep analysis. Feature-based machine learning approaches involved the extraction of handcrafted features from EEG signals in multiple domains, followed by classification using supervised learning algorithms. In the time domain, features such as statistical moments including mean, variance, skewness, and kurtosis, along with zero-crossing rates and signal energy, were commonly employed. In the frequency domain, spectral features derived from Fast Fourier Transform analysis, including absolute and relative power in clinically defined frequency bands (delta: 0.5-4 Hz, theta: 4-8 Hz, alpha: 8-13 Hz, sigma: 12-16 Hz, and beta: 13-30 Hz), provided discriminative information about different sleep states. Nonlinear features including entropy measures, fractal dimensions, and correlation dimensions were also explored to capture the complex dynamic properties of EEG signals.

Support Vector Machines (SVMs) with various kernel functions were among the most widely applied classifiers for EEG-based sleep disorder detection. SVMs demonstrated competitive performance by maximizing the margin between class boundaries in high-dimensional feature spaces, making them effective for handling the high variability in EEG data. Random Forest classifiers, which aggregate decisions from multiple independently trained decision trees, were also extensively used due to their robustness to noise and ability to handle high-dimensional feature vectors. Other classical classifiers including k-Nearest Neighbors, Naive Bayes, and Linear Discriminant Analysis were evaluated, though they generally showed inferior performance compared to SVMs and ensemble methods.

Wavelet-based feature extraction methods provided an important advancement by enabling time-frequency analysis of EEG signals. The Discrete Wavelet Transform (DWT) decomposes EEG signals into multiple frequency sub-bands at different temporal scales, capturing non-stationary characteristics that Fourier-based methods cannot represent. Wavelet coefficients from clinically relevant frequency sub-bands were used to compute statistical features that effectively characterized sleep stage-specific EEG patterns. Studies demonstrated that wavelet-based features complemented spectral features, improving classification performance when combined within multi-domain feature vectors.

Despite the progress achieved with feature-based machine learning approaches, these methods faced fundamental limitations that restricted their clinical applicability. The quality and relevance of extracted features depended heavily on domain expert knowledge and the specific characteristics of the EEG recording system, making feature engineering a time-intensive and dataset-specific process. Furthermore, the complexity and variability of EEG signals across individuals and recording conditions meant that handcrafted features often failed to generalize to new datasets, limiting the practical utility of these models.

The advent of deep learning introduced a paradigm shift in automated EEG analysis by enabling models to learn feature representations directly from raw or minimally

preprocessed data, without the need for manual feature engineering. Convolutional Neural Networks (CNNs) were among the first deep learning architectures applied to EEG-based sleep classification tasks. The convolutional layers of CNNs apply learned filters across the input signal to detect local patterns and features, providing an automated alternative to handcrafted feature extraction. Pooling layers subsequently reduce the dimensionality of extracted features while preserving the most salient information. Seminal work by Tsinalis et al. demonstrated that CNNs could learn discriminative representations from single-channel EEG signals for sleep stage classification, achieving performance comparable to traditional machine learning approaches with substantially less preprocessing effort.

DeepSleepNet, proposed by Supratak et al., represented a significant advancement in CNN-based sleep staging by combining convolutional feature extraction with sequence modeling. The model employed two parallel CNN branches with different filter sizes to capture both fine-grained and coarse temporal features from EEG epochs, followed by bidirectional LSTM layers to model the sequential dependencies between consecutive sleep stages. This hybrid CNN-LSTM architecture demonstrated state-of-the-art performance on multiple benchmark datasets, establishing the importance of integrating spatial and temporal modeling for EEG-based sleep analysis.

Attention mechanisms, originally developed for natural language processing tasks, were subsequently adapted for EEG signal analysis to address the limitations of uniform feature aggregation in CNN and RNN models. Attention-based models assign adaptive importance weights to different parts of the input signal, enabling the model to focus on the most diagnostically relevant features. Eldele et al. proposed an attention-based deep learning approach for sleep stage classification that utilized multi-resolution convolutional modules with attention mechanisms to capture EEG features at multiple temporal scales. This approach demonstrated significant improvements in classification performance compared to CNN-only and LSTM-only baselines.

The introduction of transformer-based architectures in sequence modeling tasks motivated researchers to apply self-attention mechanisms to EEG analysis.

Transformers process input sequences in parallel using self-attention, enabling efficient modeling of long-range dependencies without the sequential computation limitations of RNNs. Hybrid models combining CNN-based local feature extraction with transformer-based global context modeling have demonstrated improved performance in sleep stage classification tasks compared to either component alone.

More recent research has focused on addressing specific practical challenges in clinical sleep disorder detection, including cross-database generalization, single-channel recording analysis, and automated sleep apnea detection. Studies have explored transfer learning strategies to improve model generalization across different recording systems and patient populations. Single-channel models are particularly relevant for wearable applications, where the practicality of data acquisition is a primary concern. Automated detection of sleep apnea events from EEG and other polysomnographic signals has been pursued using diverse deep learning approaches, including CNN-LSTM hybrids and attention-based networks.

Table 2.1: Comparative Analysis of Existing Sleep Disorder Detection Techniques

Ref	Authors & Year	Journal/Conf	Method	Problem	Solution	Limitations
[4]	Supratak et al., 2017	IEEE TNSRE	DeepSleepNet (CNN+BiLSTM)	Sleep stage classification	Combines CNN and BiLSTM	Limited to single-channel; no attention
[5]	Vaswani et al., 2017	NeurIPS	Transformer / Multi-Head Attention	Sequence modeling	Global dependency modeling	High computational cost
[9]	Li et al., 2022	IEEE Access	EEGNet (CNN-BiLSTM)	Sleep stage classification	Integrates spatial & temporal features	No attention mechanism
[11]	Eldele et al., 2021	IEEE TNSRE	Attention-based CNN	Sleep stage scoring	Multi-resolution attention	Single-channel only
[12]	Chen et al., 2023	IEEE BIBM	CNN-Transformer-LSTM	Automatic sleep staging	Hybrid global-local modeling	Large model size; complex training
[13]	Acharya et al., 2023	Knowledge-Based Systems	Deep CNN	Sleep apnea detection	Automated apnea classification	Lacks temporal modeling

[10]	Michielli et al., 2019	CBM	LSTM	Sleep stage classification	Temporal dependency modeling	No spatial feature extraction
[14]	Almutairi, 2024	Sensors	Deep learning review	Sleep disorder detection review	Comprehensive survey	No novel model proposed

From Table 2.1, it can be observed that sleep disorder detection techniques have evolved significantly from traditional machine learning approaches to advanced hybrid deep learning architectures. Early approaches primarily focused on textual and statistical features and relied heavily on handcrafted representations, which limited their ability to generalize across diverse EEG recording conditions and patient populations. With the introduction of deep learning models such as CNNs and RNNs, the ability to capture local spatial patterns and temporal dependencies improved substantially.

More recent approaches have shifted towards hybrid architectures that combine multiple complementary components, such as CNN-BiLSTM and CNN-Transformer models, demonstrating improved performance by leveraging both spatial and sequential learning. However, several limitations persist. Many models still focus predominantly on either textual or structural information, failing to achieve a balanced integration of both. Additionally, existing approaches often lack robust mechanisms for handling class imbalance, which is a prevalent challenge in clinical EEG datasets. These observations highlight the need for a more comprehensive framework that effectively integrates multiple feature learning strategies while addressing practical challenges in clinical deployment.

2.2 LIMITATIONS OF EXISTING SYSTEM

Although existing sleep disorder detection methods have achieved considerable progress, they continue to face several fundamental limitations that hinder their effectiveness in real-world clinical applications. A systematic analysis of these limitations reveals the primary bottlenecks that prevent current approaches from achieving the level of performance and reliability required for clinical adoption.

One of the primary challenges is the dependence on handcrafted feature engineering in traditional machine learning approaches. Feature extraction requires substantial domain expertise and is inherently dataset-specific, limiting the generalizability of models trained on one dataset when applied to EEG recordings from different clinical settings or recording systems. The manual feature design process is time-consuming, subjective, and may fail to capture complex nonlinear patterns that are diagnostically relevant but difficult to quantify through predefined mathematical operations. Even carefully designed feature sets may not adequately represent the full complexity of EEG dynamics across all sleep disorder categories.

Another significant limitation lies in the inability of many existing models to effectively integrate spatial and temporal information from EEG signals. Standalone CNN models excel at extracting local spatial patterns from individual EEG epochs but lack mechanisms for modeling the temporal dependencies and stage transitions that span across multiple consecutive epochs. Conversely, RNN-based models capture sequential dynamics effectively but may fail to extract detailed local signal morphologies. This separation of spatial and temporal feature learning results in incomplete representations that limit overall classification performance.

Class imbalance represents a critical practical challenge in EEG-based sleep disorder detection. Clinical EEG datasets typically contain significantly more recordings from common sleep stages or normal sleep patterns compared to rare disorder-specific patterns. For example, in datasets collected from general clinical populations, normal sleep EEG segments may substantially outnumber those associated with narcolepsy or severe sleep-disordered breathing episodes. Models trained on imbalanced datasets tend to exhibit biased predictions favoring majority classes, resulting in high overall accuracy but poor sensitivity for minority disorder categories. Without appropriate strategies for addressing class imbalance, the clinical utility of automated detection systems is severely compromised.

The lack of model interpretability represents another major limitation, particularly in the context of clinical applications. Deep learning models, especially complex hybrid

architectures, often function as black-box systems that provide classification outputs without explanations of the underlying decision-making processes. This opacity is problematic in medical applications, where healthcare professionals require transparency and explainability to trust and validate automated diagnostic recommendations. The inability to identify which specific EEG features or temporal patterns contributed to a classification decision limits the adoption of deep learning systems in clinical practice.

Computational requirements present additional practical limitations for clinical deployment. Transformer-based architectures and large hybrid models require substantial computational resources for both training and inference, making real-time analysis challenging in resource-constrained environments such as wearable monitoring devices or rural healthcare settings. The trade-off between model complexity and computational efficiency remains an important consideration for practical deployment.

2.3 GAPS IDENTIFIED

The analysis of existing sleep disorder detection techniques highlights several critical gaps that restrict their effectiveness in real-world clinical environments. Although significant progress has been made through the use of machine learning and deep learning approaches, these methods are often developed under controlled conditions that do not fully reflect the complexity and variability of clinical EEG data.

2.3.1 Lack of Effective Hybrid Modelling

Most existing approaches to EEG-based sleep disorder detection tend to focus on either spatial feature extraction or temporal sequence modelling independently, rather than combining both aspects within a unified framework. Spatial models such as CNNs are effective at capturing local signal patterns within individual EEG epochs, while temporal models like RNNs excel in learning sequential dependencies across consecutive time windows. However, the absence of proper integration between these two complementary learning strategies results in incomplete feature representation.

Sleep disorders manifest through complex spatiotemporal patterns in EEG signals that require simultaneous analysis of local morphological features and long-range temporal dynamics. Models that address only one dimension of this complexity inherently miss important diagnostic information, limiting their classification accuracy and clinical relevance.

2.3.2 Limited Use of Attention Mechanisms

Attention mechanisms have demonstrated significant improvements across diverse deep learning applications by enabling models to selectively focus on the most relevant parts of the input data during prediction. Despite their demonstrated effectiveness, their application in EEG-based sleep disorder detection remains limited and underexplored. Existing models often apply uniform aggregation across all time steps and EEG channels without explicitly identifying the most diagnostically informative segments or electrode locations. This approach may include irrelevant or noisy signal segments in the feature computation while potentially underweighting the most critical EEG patterns. The incorporation of attention mechanisms could enable models to automatically identify and prioritize the EEG features most relevant to each disorder category, improving both classification performance and interpretability.

2.3.3 Poor Handling of Class Imbalance

Class imbalance is a pervasive challenge in clinical EEG datasets for sleep disorder detection. The natural prevalence rates of different sleep disorders in clinical populations result in highly skewed training datasets, where common conditions are heavily overrepresented relative to rarer disorders. Many existing systems do not adequately address this fundamental challenge, instead relying on accuracy-based optimization that inadvertently favors the majority class. Models trained without explicit imbalance handling tend to achieve high overall accuracy while demonstrating poor sensitivity for minority disorder categories, which are often the most clinically important cases. Effective strategies for addressing class imbalance, including oversampling of minority classes, class-weighted loss functions, and threshold optimization, are essential for developing clinically useful automated detection systems.

2.3.4 Lack of Interpretability in Deep Models

Deep learning models used in EEG analysis frequently function as black-box systems, generating predictions without providing human-interpretable explanations of the underlying decision processes. This fundamental lack of transparency poses significant challenges for clinical applications, where understanding the reasoning behind automated diagnostic recommendations is essential for clinical acceptance and regulatory approval. Without interpretable outputs, clinical experts cannot validate the physiological basis of automated predictions or identify cases where the model may be relying on spurious correlations rather than genuine biomarkers. The development of inherently interpretable architectures or post-hoc explanation methods that can reveal the EEG features driving classification decisions is a critical gap that must be addressed to enable clinical translation of automated sleep disorder detection systems.

2.3.5 High Computational Requirements

Advanced deep learning architectures, particularly transformer-based models and large hybrid networks, require substantial computational resources for both training and inference. These high computational demands present practical barriers to clinical deployment, especially in settings where real-time analysis is required or computational resources are limited. Memory requirements for processing long EEG sequences within a single forward pass can exceed the capacity of standard clinical hardware. Inference latency limitations may prevent the use of complex models in applications requiring real-time feedback. The development of computationally efficient architectures that maintain high classification performance while operating within practical resource constraints represents an important open challenge for the field.

The gaps identified in existing sleep disorder detection approaches clearly demonstrate that current models are not fully equipped to address the complex challenges of real-world clinical EEG analysis. The need for effective hybrid modeling, improved use of attention mechanisms, robust class imbalance handling, enhanced interpretability, and computationally efficient architectures motivates the development of the proposed framework presented in this work.

2.4 PROBLEM STATEMENT

Sleep disorders such as insomnia, narcolepsy, and sleep-disordered breathing significantly impact human health and quality of life, requiring accurate and timely diagnosis for effective treatment. Traditional diagnostic methods such as polysomnography are expensive, time-consuming, and require expert clinical analysis, making them unsuitable for large-scale screening or continuous monitoring applications. Existing automated approaches either rely on handcrafted feature engineering with limited generalization or fail to effectively integrate both spatial and temporal characteristics of EEG signals within a unified learning framework. Additionally, issues such as class imbalance, lack of interpretability, and variability in EEG data across subjects and recording conditions further limit the performance and clinical applicability of current systems. Therefore, the problem addressed in this work is to develop an efficient, reliable, and interpretable deep learning-based system for multi-class sleep disorder detection using EEG signals, capable of capturing complex spatiotemporal patterns while addressing practical challenges including class imbalance and data variability, to produce accurate predictions suitable for real-world clinical deployment.

2.5 OBJECTIVES

The objectives of the proposed method are:

- i. To develop a hybrid deep learning model integrating CNN, BiLSTM, and Multi-Head Self-Attention mechanisms for accurate multi-class sleep disorder detection from EEG signals.
- ii. To design an efficient EEG signal preprocessing and multi-domain feature extraction pipeline that enhances signal representation and supports robust model learning.
- iii. To build a reliable and scalable automated system that handles class imbalance, provides interpretable predictions through attention mechanisms, and improves diagnostic performance for real-world clinical applications.

CHAPTER 3 PROPOSED SYSTEM

3.1 OVERVIEW OF PROPOSED SYSTEM

The increasing prevalence of sleep disorders and their significant impact on public health necessitates the development of accurate, efficient, and clinically practical automated detection systems. The limitations identified in existing approaches, including the inability to effectively integrate spatial and temporal EEG features, inadequate handling of class imbalance, and lack of model interpretability, motivate the development of a comprehensive hybrid deep learning framework that addresses these challenges within a unified architecture.

In this work, we propose a deep learning-based framework for automated sleep disorder detection using electroencephalography (EEG) signals. The system is designed to effectively classify multiple sleep disorders, including insomnia, narcolepsy, and sleep-disordered breathing, under varying data conditions and realistic clinical constraints. Unlike traditional approaches that rely heavily on manual feature extraction, the proposed system leverages a hybrid deep learning architecture to automatically learn meaningful, hierarchical representations from EEG data.

The proposed framework integrates three complementary learning components: Convolutional Neural Networks (CNN) for spatial feature extraction, Bidirectional Long Short-Term Memory (BiLSTM) networks for temporal sequence modeling, and Multi-Head Self-Attention mechanisms for adaptive feature weighting and interpretability. CNN layers capture local spatial patterns and morphological features within EEG epochs, identifying characteristic waveform signatures associated with different sleep states. The BiLSTM component processes the sequential output of the CNN layers in both forward and backward temporal directions, enabling comprehensive modeling of stage transitions and long-range temporal dependencies. The Multi-Head Self-Attention mechanism further enhances the model by computing context-dependent importance weights that identify the most diagnostically relevant features and temporal segments.

The system follows a structured end-to-end pipeline that begins with EEG signal acquisition and proceeds through preprocessing, feature extraction, model training, and classification. The preprocessing stage addresses signal quality through bandpass filtering, artifact removal, and epoch segmentation. Feature extraction enhances signal representation by computing multi-domain features including spectral, statistical, and wavelet-based characteristics. By combining these components within a unified framework, the proposed system aims to achieve high classification accuracy, robust generalization across subjects and recording conditions, and clinically meaningful interpretability.

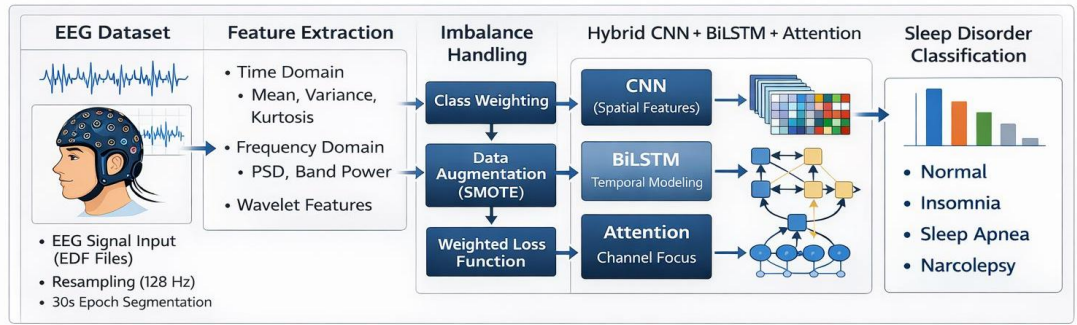


Fig 3.1: Overview of the Proposed Hybrid CNN-BiLSTM Architecture

Figure 3.1 illustrates the overall architecture of the proposed EEG-based sleep disorder detection framework. The model processes EEG signals through a structured pipeline beginning with preprocessing and feature extraction, followed by class imbalance handling and hybrid deep learning. The CNN component extracts spatial features from the preprocessed EEG epochs, while the BiLSTM captures temporal dependencies across the sequence. The Multi-Head Self-Attention mechanism selectively weights features before the final classification layer assigns input samples to one of the four disorder categories.

3.2 SYSTEM ARCHITECTURE

The system architecture of the proposed framework defines the structural design and operational flow of the model for sleep disorder detection from EEG signals. The architecture is designed to address key challenges including class imbalance, complex

temporal dependencies, and the need for interpretable predictions through the systematic integration of multiple complementary learning components.

The architecture begins with the input stage, where raw EEG recordings are received in standard clinical formats such as European Data Format (EDF). Each recording contains multichannel EEG signals that are processed through the preprocessing pipeline to produce clean, standardized epoch segments suitable for deep learning analysis. The preprocessed epochs serve as the primary input to the hybrid neural network model.

The first major processing component is the CNN module, which consists of multiple convolutional layers with increasing filter depths and decreasing spatial dimensions through pooling operations. Each convolutional layer applies a set of learned filters across the input EEG epoch, detecting local temporal patterns at different scales and frequencies. Batch normalization is applied after each convolutional layer to stabilize training and improve convergence speed. Dropout regularization is incorporated to prevent overfitting by randomly deactivating a fraction of neural connections during training.

The output of the CNN module is fed into the BiLSTM component, which processes the spatial feature sequences in both forward and backward temporal directions. The forward LSTM processes the sequence from the first time step to the last, building a temporal context that incorporates information from preceding time steps. The backward LSTM processes the same sequence in reverse order, incorporating contextual information from future time steps. The outputs of the forward and backward LSTMs are concatenated at each time step to produce a bidirectional representation that captures temporal context from both directions simultaneously.

Following the BiLSTM component, the Multi-Head Self-Attention mechanism computes context-dependent importance weights across the temporal sequence. The attention module divides the representation into multiple parallel attention heads, each of which independently computes attention weights over the sequence. This multi-head

structure allows the model to attend to different aspects of the temporal dependencies simultaneously, capturing diverse diagnostic patterns at different temporal scales. The outputs of all attention heads are concatenated and linearly projected to produce the final attention-weighted representation.

The final classification stage consists of fully connected layers with dropout regularization, followed by a softmax output layer that produces probability distributions over the four sleep disorder classes: Normal, Insomnia, Narcolepsy, and Sleep Disordered Breathing. The predicted class is determined by the maximum probability output from the softmax layer.

3.3 PROBLEM FORMULATION

The sleep disorder detection task is formulated as a multi-class classification problem based on EEG signal data. Let the EEG dataset be represented as:

$$X = \{x_1, x_2, x_3, \dots, x_n\} \quad 3.1$$

where each x_i represents an EEG epoch corresponding to a fixed time segment of T seconds, sampled at frequency f_s Hz, resulting in a signal vector of length $T \times f_s$ per channel.

Each sample is associated with a ground truth label:

$$Y = \{y_1, y_2, y_3, \dots, y_n\} \quad 3.2$$

where y_i belongs to one of the predefined classes $C = \{\text{Normal, Insomnia, Narcolepsy, Sleep Disordered Breathing}\}$. The objective of the model is to learn a function:

$$f(X) \rightarrow Y \quad 3.3$$

that maps EEG signal epochs to their corresponding sleep disorder classes. The learning problem is formulated as the minimization of a class-weighted cross-entropy loss function to address the inherent class imbalance in the training data.

The dataset is partitioned into training set D_{train} for model learning, validation set D_{val} for hyperparameter optimization and early stopping, and test set D_{test} for final performance evaluation. The partitioning is performed at the subject level to prevent data leakage and ensure that the model's performance on the test set reflects its generalization to unseen patients.

3.4 EEG PRE-PROCESSING AND FEATURE EXTRACTION

EEG preprocessing is a crucial step in ensuring signal quality and consistency for downstream deep learning analysis. Raw EEG signals collected in clinical settings are frequently contaminated by artifacts arising from eye movements, muscle contractions, electrical interference, and electrode impedance variations. The preprocessing pipeline is designed to systematically remove these artifacts while preserving the diagnostically relevant signal components.

The preprocessing pipeline consists of several sequential operations. Bandpass filtering is applied first, using a zero-phase digital filter with a passband of 0.5 to 30 Hz to remove low-frequency drift artifacts below 0.5 Hz and high-frequency muscle artifacts and electrical noise above 30 Hz. This frequency range encompasses the clinically relevant EEG frequency bands including delta, theta, alpha, sigma, and beta, while effectively excluding the most common artifact sources. Resampling is performed following filtering to standardize all EEG recordings to a common sampling frequency, ensuring uniformity of input dimensions across the dataset. Epoch segmentation divides the continuous EEG recording into fixed-length segments of 30 seconds, consistent with clinical sleep scoring conventions. Artifacts within individual epochs are detected using amplitude thresholding and rejected to maintain data quality.

Following preprocessing, multi-domain feature extraction is performed to enhance the representation of EEG signals beyond the raw waveform. Statistical features capture the distributional properties of the EEG signal within each epoch, including mean amplitude, standard deviation, skewness, kurtosis, and root mean square values. These

features provide a compact summary of signal amplitude characteristics and have demonstrated discriminative value in distinguishing between sleep stages. Spectral features are derived from Welch's power spectral density estimation, providing frequency-domain characterization of the EEG signal. Band power features are computed for the clinically defined frequency bands, capturing the relative contribution of each frequency component to the overall signal power. Wavelet-based features are computed using discrete wavelet decomposition at multiple resolution levels, providing time-frequency analysis that captures the non-stationary dynamics of EEG signals across sleep stage transitions.

3.5 HYBRID MODEL DESIGN (CNN-BiLSTM-ATTENTION)

The proposed hybrid model architecture combines three complementary neural network components within a unified end-to-end trainable framework. Each component is designed to capture specific aspects of the complex spatiotemporal structure of EEG sleep signals.

The CNN component serves as the spatial feature extractor, processing individual EEG epochs to identify local temporal patterns and waveform morphologies characteristic of different sleep states. The architecture consists of three convolutional blocks, each containing a convolutional layer, batch normalization, ReLU activation, and max pooling. The first convolutional block employs 32 filters of kernel size 7 to capture broad temporal patterns, while subsequent blocks use 64 and 128 filters with progressively smaller kernels of sizes 5 and 3, respectively, to detect finer-grained features. This hierarchical feature extraction structure enables the CNN to learn representations at multiple temporal scales, from broad sleep stage morphologies to fine-grained waveform details.

The BiLSTM component models the temporal dependencies in the sequence of CNN-extracted feature vectors. The architecture consists of two stacked BiLSTM layers with hidden dimensions of 128 and 64, respectively. Each BiLSTM layer

processes the input sequence using parallel forward and backward LSTM cells, with their outputs concatenated at each time step to produce a bidirectional representation. Dropout with a rate of 0.3 is applied to the outputs of each BiLSTM layer to regularize the model and prevent overfitting to training data. The stacked BiLSTM structure allows the model to learn hierarchical temporal representations, with lower layers capturing local temporal dependencies and higher layers modeling longer-range contextual patterns.

The Multi-Head Self-Attention mechanism is applied to the BiLSTM output sequence to compute context-dependent importance weights. The attention module uses 8 parallel attention heads, each computing scaled dot-product attention independently over the temporal sequence. The attention computation for each head h is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) V \quad 3.4$$

where Q , K , and V are query, key, and value matrices derived from linear projections of the BiLSTM output, and d_k is the dimension of the key vectors. The outputs of all 8 attention heads are concatenated and linearly projected to produce the final attended representation. This multi-head structure enables the model to capture diverse temporal relationships simultaneously, attending to different aspects of the EEG sequence for each diagnostic pattern.

3.6 CLASS IMBALANCE HANDLING AND TRAINING STRATEGY

Addressing the class imbalance in clinical EEG datasets is essential for developing models with practical diagnostic utility. The proposed system incorporates multiple complementary strategies for handling class imbalance at both the data level and the algorithmic level.

At the data level, the Synthetic Minority Over-sampling Technique (SMOTE) is applied to generate synthetic training samples for underrepresented disorder categories. SMOTE creates new synthetic samples by interpolating between existing minority class

instances in the feature space, effectively increasing the representation of rare disorder categories without simply duplicating existing samples. This data-level augmentation approach helps the model learn more robust decision boundaries for minority classes without introducing exact duplicates that might lead to overfitting.

At the algorithmic level, a class-weighted cross-entropy loss function is employed during model training. The loss weights for each class are computed as the inverse of the class frequency in the training dataset, normalized to ensure that the overall loss magnitude remains stable. This weighting scheme penalizes misclassification of minority class samples more heavily than majority class errors, encouraging the model to develop sensitivity for all disorder categories regardless of their prevalence in the training data.

The model is trained using the Adam optimizer with an initial learning rate of 0.001, weight decay of $1e-4$, and an 80-10-10 train-validation-test split. A learning rate scheduler reduces the learning rate by a factor of 0.5 when the validation loss plateaus for 5 consecutive epochs, enabling fine-grained optimization as training progresses. Early stopping with a patience of 15 epochs is employed to prevent overfitting, with model checkpointing preserving the weights corresponding to the best validation performance.

3.7 MODEL INTERPRETABILITY AND ADVANTAGES

One of the key contributions of the proposed system is the integration of Multi-Head Self-Attention mechanisms that provide inherent interpretability alongside improved classification performance. The attention weights computed by the attention module during inference can be visualized to identify the temporal segments and EEG features that contributed most significantly to the classification decision for each input sample. This interpretability mechanism enables clinical experts to validate the physiological

basis of automated predictions and provides insights into the EEG patterns characteristic of each disorder category.

The attention weight maps can be visualized as heatmaps over the temporal dimension of the EEG sequence, highlighting the time points that received the highest attention weights for each prediction. These visualizations enable qualitative assessment of whether the model is attending to clinically meaningful EEG features, such as specific waveform events or frequency band characteristics associated with known disorder biomarkers. The multi-head structure further enables visualization of diverse attention patterns, revealing the different aspects of the EEG sequence that each attention head focuses on.

The advantages of the proposed system include effective integration of spatial and temporal features through the CNN-BiLSTM hybrid architecture, improved classification accuracy compared to standalone models, robust handling of class imbalance through combined data augmentation and algorithmic weighting strategies, enhanced interpretability through attention visualization, and scalability for real-world clinical applications through the use of standard deep learning frameworks and hardware.

CHAPTER 4 SYSTEM DESIGN

4.1 OVERVIEW OF SYSTEM DESIGN

The system design of the proposed framework focuses on developing a robust, scalable, and clinically practical solution for automated sleep disorder detection using EEG signals. The design integrates signal preprocessing, multi-domain feature extraction, and hybrid deep learning within a structured pipeline that addresses the fundamental challenges of EEG-based clinical diagnosis.

In clinical sleep monitoring environments, EEG data is acquired using standardized electrode placement systems across multiple channels. The resulting multichannel recordings contain rich spatial and temporal information about brain electrical activity that reflects the dynamic transitions between sleep stages and the pathological disruptions associated with different disorder types. The system design must therefore accommodate the inherent complexity of multichannel EEG data while maintaining computational efficiency and generalization capability.

The system design focuses on effectively capturing the spatiotemporal characteristics of EEG signals through the combined application of spatial feature extraction via CNN layers, bidirectional temporal modeling via BiLSTM networks, and adaptive feature weighting via Multi-Head Self-Attention mechanisms. The integration of attention mechanisms into the architecture design enhances both the classification performance and the interpretability of the system, providing clinically meaningful insights into the EEG features driving automated diagnostic decisions.

Overall, the system is designed to provide a robust and reliable solution for multi-class sleep disorder detection, supporting its integration into clinical decision support workflows and enabling consistent, objective diagnostic assistance across diverse patient populations and clinical settings.

4.2 WORKFLOW OF PROPOSED SYSTEM



Fig 4.1: EEG Signal Preprocessing and Feature Extraction Pipeline

The workflow of the proposed system follows a systematic and modular sequence of operations, starting from raw EEG signal acquisition through final sleep disorder classification. Each stage of the workflow is designed to prepare the data progressively for the subsequent stage, ensuring that the input to the deep learning model represents a high-quality, standardized, and informative representation of the original clinical EEG recordings.

The workflow begins with EEG signal acquisition, where raw EEG data is collected from clinical sleep recordings in standard formats such as the European Data Format (EDF). These signals represent the electrical activity of the brain during sleep and serve as the primary input for the automated detection system. The acquired signals typically contain multiple EEG channels recorded simultaneously, along with metadata such as recording date, patient information, and sleep stage annotations provided by clinical experts.

In the preprocessing stage, the raw EEG signals undergo a series of processing operations to remove artifacts, standardize signal properties, and prepare epochs for feature extraction. Bandpass filtering with a passband of 0.5 to 30 Hz removes low-frequency drift and high-frequency noise while preserving diagnostically relevant EEG components. Resampling standardizes the sampling frequency across all recordings, ensuring uniform input dimensions. Epoch segmentation divides the continuous EEG recording into fixed-length segments aligned with clinical sleep scoring conventions.

Artifact detection and rejection removes segments contaminated by excessive noise from ocular movements, muscle activity, or electrode artifacts.

Following preprocessing, the system performs multi-domain feature extraction from each EEG epoch. Statistical features capture amplitude and distributional properties of the signal. Spectral features provide frequency-domain characterization through power spectral density estimation. Wavelet-based features capture time-frequency characteristics that reflect the non-stationary dynamics of EEG during sleep. These complementary feature types are concatenated to form a comprehensive feature vector that serves as an enriched input representation for the hybrid deep learning model.

The extracted feature vectors are then fed into the hybrid deep learning model, which consists of sequentially arranged CNN, BiLSTM, and Multi-Head Self-Attention components. The CNN processes each epoch's feature vector to extract spatial representations, which are then organized into temporal sequences and processed by the BiLSTM to capture sequential dependencies across consecutive epochs. The attention mechanism weights the temporal features adaptively before the final classification layer assigns predictions.

The training workflow incorporates class imbalance handling through SMOTE oversampling and class-weighted loss computation. Model performance is monitored on the validation set after each training epoch, with early stopping and learning rate scheduling applied to optimize training convergence. The final trained model is evaluated on the held-out test set using standard performance metrics including accuracy, precision, recall, and F1-score.

The inference workflow for new EEG recordings follows the same preprocessing and feature extraction pipeline used during training, ensuring consistency between training and deployment conditions. The trained model processes the prepared feature sequences and produces classification predictions for each epoch, which can be aggregated to provide disorder-level diagnoses for entire recordings.

4.3 EEG SIGNAL REPRESENTATION

In the proposed system, EEG signals are represented in a structured format that captures both temporal and spectral characteristics essential for sleep disorder classification. The transformation from raw clinical EEG recordings to the structured representation used by the deep learning model involves multiple stages of signal processing and feature computation.

Raw EEG recordings are multichannel time-series signals measured at standardized electrode locations according to the international 10-20 system. Each channel captures the electrical potential difference between an active scalp electrode and a common reference electrode, providing spatial coverage of different brain regions. The temporal resolution of EEG recordings is determined by the sampling frequency, which is standardized to 128 Hz after resampling in the preprocessing pipeline.

EEG recordings are divided into fixed-length epochs of 30 seconds, consistent with clinical sleep scoring conventions established by sleep medicine organizations. Each epoch contains $30 \times 128 = 3,840$ samples per channel, providing a structured temporal window for feature extraction and classification. The 30-second epoch duration was chosen to align with the established clinical convention for sleep stage scoring, where each epoch is assigned a single sleep stage label by expert clinicians.

The EEG signal representation incorporates both raw waveform data and computed feature vectors. Raw waveform data preserves the full temporal resolution of the EEG signal, enabling the CNN component to identify fine-grained local patterns. Computed feature vectors provide a complementary representation that captures frequency-domain, statistical, and time-frequency characteristics that may not be directly apparent in the raw waveform. The combined representation provides the deep learning model with comprehensive information about each EEG epoch from multiple analytical perspectives.

The representation of EEG signals includes time-domain features that describe statistical properties of signal amplitude, frequency-domain features that characterize spectral content in clinically defined bands, and wavelet-based features that capture non-stationary time-frequency dynamics. This multi-dimensional representation allows the model to learn complex patterns associated with different sleep disorders, improving both sensitivity and specificity of classification across all disorder categories.

4.4 MODEL DESIGN

The detailed model design specifies the architectural parameters, layer configurations, and training procedures that implement the proposed hybrid CNN-BiLSTM-Attention framework for sleep disorder detection.

The input layer receives preprocessed EEG epoch feature vectors of dimension d_{input} , where d_{input} is determined by the number of extracted features per epoch. The input is organized as a sequence of feature vectors, where each element of the sequence corresponds to one EEG epoch and the sequence length corresponds to the number of consecutive epochs in the input window.

The CNN module processes individual epochs using three convolutional blocks with the following architecture. The first block employs 32 convolutional filters of kernel size 7, followed by batch normalization, ReLU activation, and max pooling with stride 2. The second block employs 64 filters of kernel size 5, followed by batch normalization, ReLU activation, and max pooling with stride 2. The third block employs 128 filters of kernel size 3, followed by batch normalization and ReLU activation. The output of the CNN module is a feature representation of reduced temporal dimension that captures multi-scale spatial patterns.

The BiLSTM module consists of two stacked BiLSTM layers. The first BiLSTM layer contains 128 hidden units in each direction (forward and backward), producing a concatenated output of dimension 256 per time step. The second BiLSTM layer

contains 64 hidden units in each direction, producing a concatenated output of dimension 128 per time step. Dropout with rate 0.3 is applied after each BiLSTM layer during training. The stacked structure enables hierarchical temporal modeling with increasing abstraction at higher layers.

The Multi-Head Self-Attention module applies 8-head attention to the BiLSTM output sequence. Each attention head uses a key and query dimension of 16 and a value dimension of 16, resulting in a concatenated output dimension of 128. A learned linear projection maps the concatenated attention output to the final attended representation. Position-wise feed-forward layers with residual connections and layer normalization are applied after the attention computation to enhance representational capacity.

The classification head consists of two fully connected layers. The first layer transforms the attention output to a hidden dimension of 64 using ReLU activation and dropout with rate 0.5. The second layer is a linear projection to the number of disorder classes (4), followed by a softmax activation that produces the final class probability distribution. The predicted class is the argument of maximum probability.

4.5 FEATURE PROCESSING AND OPTIMIZATION

Feature processing plays a fundamental role in ensuring that the input representations provided to the deep learning model are consistent, informative, and numerically stable. The extracted multi-domain features undergo normalization and standardization procedures to ensure compatibility with gradient-based optimization and to prevent features with large numerical ranges from dominating the learning process.

Z-score normalization is applied to all extracted features, transforming each feature dimension to have zero mean and unit standard deviation based on statistics computed from the training dataset. This normalization ensures that all feature dimensions

contribute on a comparable scale to the model's learning, preventing features with inherently large magnitudes from receiving disproportionate gradient updates during backpropagation. The normalization parameters are computed exclusively from the training set and applied identically to the validation and test sets to prevent data leakage.

Principal Component Analysis (PCA) is applied as an optional dimensionality reduction step for high-dimensional feature vectors, reducing computational requirements while preserving the most informative feature dimensions. The number of principal components is selected to retain 95% of the total variance in the training feature set. PCA reduces the risk of the curse of dimensionality and can improve model generalization by removing redundant and noisy feature dimensions.

Optimization techniques applied during training include the Adam optimizer with adaptive learning rates, gradient clipping to prevent exploding gradients in the BiLSTM component, learning rate scheduling based on validation performance plateaus, and L2 weight regularization to prevent overfitting. These optimization strategies collectively contribute to stable, efficient training convergence and improved model generalization.

4.6 VALIDATION AND RELIABILITY

To ensure the reliability and clinical validity of the proposed system, a rigorous validation framework is implemented that evaluates model performance under realistic conditions. The validation strategy is designed to assess both the statistical performance and the practical applicability of the automated detection system.

Subject-level data partitioning is employed to prevent data leakage between training and evaluation sets. By ensuring that all EEG epochs from a given subject appear exclusively in one partition (training, validation, or test), the evaluation provides an unbiased estimate of the model's ability to generalize to new patients rather than memorizing subject-specific patterns. This subject-independent evaluation is essential for assessing clinical applicability.

Performance is evaluated using multiple complementary metrics to provide a comprehensive assessment of classification capability. Accuracy measures the overall proportion of correct predictions across all classes. Precision evaluates the reliability of positive predictions for each class. Recall assesses the model's sensitivity for identifying each disorder category. The macro-averaged F1-score provides a balanced evaluation that weights all classes equally, regardless of their frequency in the test set. This metric is particularly important for clinical applications where performance on minority classes is as important as performance on majority classes.

Confidence calibration is assessed to verify that the model's probability outputs accurately reflect empirical prediction frequencies. Well-calibrated confidence scores enable clinical decision support systems to communicate meaningful uncertainty estimates to healthcare providers, supporting risk-stratified clinical decision-making.

4.7 DESIGN ADVANTAGES

The proposed system design offers several key advantages over traditional and contemporary approaches to automated sleep disorder detection. These advantages collectively make the system more suitable for real-world clinical deployment than existing alternatives.

The hybrid CNN-BiLSTM-Attention architecture effectively integrates spatial and temporal feature extraction within a unified end-to-end trainable framework, enabling comprehensive analysis of EEG signals that captures both local morphological patterns and long-range temporal dependencies. This integration addresses the fundamental limitation of standalone models that focus exclusively on one dimension of EEG complexity.

The incorporation of Multi-Head Self-Attention mechanisms provides inherent model interpretability through attention weight visualization, enabling clinical experts

to validate the physiological basis of automated predictions. This interpretability is essential for clinical trust and regulatory acceptance of automated diagnostic systems.

The combined use of SMOTE oversampling and class-weighted loss functions addresses class imbalance at both the data and algorithmic levels, ensuring balanced performance across all disorder categories rather than optimizing exclusively for majority class accuracy. This balanced approach is clinically essential for detecting rare but serious conditions.

The modular pipeline architecture enables flexible adaptation to different EEG recording systems, clinical protocols, and computational constraints. Individual pipeline components can be updated or replaced independently without disrupting the overall system, facilitating maintenance and improvement over time. The use of standard deep learning frameworks and hardware ensures broad compatibility with clinical computing infrastructure.

CHAPTER 5 IMPLEMENTATION AND TESTING

5.1 IMPLEMENTATION OVERVIEW

The proposed system was implemented using Python, leveraging advanced deep learning and signal processing libraries for EEG analysis and classification. The implementation focuses on building a complete, modular pipeline that encompasses EEG data loading, preprocessing, feature extraction, model design, training, and evaluation. Each component is implemented as an independent module with well-defined interfaces, enabling flexible configuration and systematic experimentation with different architectural choices and hyperparameter settings.

The development environment employs PyTorch as the primary deep learning framework, chosen for its dynamic computational graph capabilities, extensive library support, and efficient GPU acceleration. EEG signal processing is performed using the MNE-Python library, which provides comprehensive tools for loading, preprocessing, and analyzing EEG data in standard clinical formats. Scientific computing is supported by NumPy and SciPy, while data management and analysis are handled using Pandas. Model performance visualization employs Matplotlib and Seaborn.

The implementation is organized into modular components including a data loading module for reading EDF-format EEG files and organizing recordings into structured datasets, a preprocessing module implementing the filtering, resampling, and segmentation pipeline, a feature extraction module computing statistical, spectral, and wavelet-based features, a model module defining the CNN-BiLSTM-Attention architecture, a training module implementing the optimization procedure with imbalance handling, and an evaluation module computing and reporting performance metrics.

5.2 DATASET DESCRIPTION

The experiments were conducted using the CAP Sleep Dataset, which is a publicly available repository of clinical polysomnographic recordings from patients with various sleep disorders. The dataset was selected for its clinical relevance, diversity of disorder categories, and compatibility with the proposed multi-class classification framework.

Table 5.1: CAP Sleep Dataset Statistics

Class	Number of Recordings	Total Epochs	Average Duration (hrs)
Normal	4	3,840	8.2
Insomnia	4	3,520	7.7
Sleep Disordered Breathing (SDB)	4	4,120	8.8
Narcolepsy	4	3,280	7.1

The CAP Sleep dataset includes full-night polysomnographic recordings from patients diagnosed with specific sleep disorders and normal control subjects. EEG signals were recorded using standardized electrode placements with a minimum sampling frequency of 128 Hz, providing adequate temporal resolution for automated analysis. Each recording is accompanied by expert clinical annotations providing ground truth sleep stage labels for each 30-second epoch.

The distribution of disorder categories in the dataset reflects the class imbalance inherent in clinical sleep disorder populations. The dataset includes 16 recordings distributed across four disorder categories, with each category represented by 4 recordings. While this provides a balanced inter-class distribution at the recording level, within-recording epoch distributions vary based on disease severity and individual sleep architecture patterns.

The EEG signals in the dataset contain multiple channels recorded from frontal, central, occipital, and temporal electrode locations, providing comprehensive spatial

coverage of brain electrical activity during sleep. The multichannel EEG data enables the model to leverage spatial patterns across different brain regions, in addition to the temporal dynamics captured within individual channels.

5.3 DATA PREPROCESSING AND FEATURE EXTRACTION

The data preprocessing and feature extraction pipeline plays a fundamental role in transforming raw clinical EEG recordings into structured, informative representations suitable for deep learning analysis. The implementation encompasses multiple sequential processing stages that are applied consistently to all recordings.

5.3.1 Data Loading and Structuring

The CAP Sleep dataset recordings are loaded from European Data Format (EDF) files using the MNE-Python library. Each EDF file contains continuous multichannel EEG recordings along with channel metadata including electrode names, sampling frequencies, and physical measurement units. The loading procedure standardizes channel selection across all recordings by selecting a consistent set of EEG channels present in all files, discarding auxiliary channels containing non-EEG physiological signals.

Ground truth sleep disorder labels are loaded from accompanying annotation files that specify the disorder category for each recording. These subject-level labels are associated with all epochs derived from the corresponding recording. The dataset is organized into a structured format where each entry contains the preprocessed EEG epoch data, the corresponding disorder class label, the subject identifier, and the epoch index within the recording.

5.3.2 Signal Preprocessing

Signal preprocessing is implemented as a sequential pipeline applied to each EEG recording before epoch extraction. Bandpass filtering is applied using a finite

impulse response (FIR) filter with a passband of 0.5 to 30 Hz, implemented in MNE-Python using a zero-phase filtering approach. Zero-phase filtering prevents temporal distortion of the filtered signal by processing the signal in both the forward and backward temporal directions, ensuring that the phase relationships between different frequency components are preserved.

Resampling is performed to standardize the sampling frequency of all recordings to 128 Hz, ensuring uniform input dimensions across the dataset. The resampling procedure uses anti-aliasing filtering to prevent aliasing artifacts that would otherwise occur when downsampling high-frequency recordings. Following resampling, the continuous EEG recording is segmented into non-overlapping 30-second epochs of 3,840 samples per channel.

Artifact detection and rejection is performed using amplitude thresholding, where epochs containing EEG amplitude values exceeding ± 150 microvolts are flagged as artifact-contaminated and excluded from the feature extraction stage. This threshold was chosen based on clinical EEG inspection guidelines to effectively remove epochs dominated by muscle artifacts, electrode pop artifacts, and other large-amplitude non-cerebral signals while retaining valid EEG epochs for analysis.

5.3.3 Feature Extraction

Multi-domain feature extraction is performed for each artifact-free EEG epoch to compute a comprehensive feature vector capturing statistical, spectral, and time-frequency characteristics of the signal. The feature extraction pipeline is applied independently to each EEG channel, with features subsequently concatenated across channels to produce a combined multichannel feature vector.

Statistical features are computed directly from the epoch time-series values. For each channel, the following statistical measures are extracted: mean amplitude, standard deviation, variance, skewness, kurtosis, peak-to-peak amplitude, root mean square, and zero-crossing rate. These features collectively characterize the amplitude

distribution and signal dynamics within each epoch, providing discriminative information about sleep stage-specific EEG properties.

Spectral features are derived from Welch's power spectral density estimation, which computes the average power spectrum from overlapping windowed segments of the EEG epoch. Absolute and relative power values are computed for the clinically defined EEG frequency bands: delta (0.5-4 Hz), theta (4-8 Hz), alpha (8-13 Hz), sigma (12-16 Hz), and beta (13-30 Hz). Spectral edge frequency and spectral centroid are computed as additional spectral shape descriptors. The ratio of power in different frequency bands is also computed, as these ratios have demonstrated discriminative value for sleep stage classification.

Wavelet-based features are computed using Daubechies 4 (db4) wavelet decomposition at 5 decomposition levels, providing time-frequency analysis at multiple resolution scales corresponding to the clinically relevant EEG frequency bands. Statistical features including mean, standard deviation, and energy are computed from the wavelet coefficients at each decomposition level, producing a compact representation of the time-frequency characteristics of the EEG signal. Wavelet entropy is additionally computed as a measure of signal complexity and regularity.

5.3.4 Label and Data Preparation

The disorder class labels are encoded as integer values for use as training targets. The four disorder categories are encoded as: Normal = 0, Insomnia = 1, Narcolepsy = 2, Sleep Disordered Breathing = 3. These integer class labels are used directly in the PyTorch cross-entropy loss computation.

Subject-level data partitioning is applied to create training, validation, and test subsets with subject independence. The 16 available subjects are randomly assigned to the three partitions in an 80-10-10 ratio at the subject level: 13 subjects for training, 2 for validation, and 1 for testing. This partitioning ensures that no subject's

recordings appear in multiple partitions, providing an unbiased assessment of generalization to new patients.

Class imbalance in the training partition is addressed through SMOTE oversampling, which generates synthetic minority class samples by interpolating between existing training examples in the feature space. The oversampling is applied exclusively to the training partition to prevent information leakage from synthetic samples into the evaluation sets. The final training dataset after SMOTE oversampling contains balanced representation across all four disorder categories.

5.3.5 Tensor and Device Management

All data components are converted into PyTorch tensor representations to enable efficient computation within the deep learning framework. The multi-domain feature matrices extracted from EEG epochs are converted to 32-bit floating point tensors, while integer class labels are stored as long integer tensors compatible with PyTorch's cross-entropy loss implementation. Graph structures representing inter-epoch temporal relationships are encoded as edge index tensors for efficient processing.

Computation device management is implemented to leverage GPU acceleration when available. The system automatically detects CUDA-compatible GPU devices and transfers all model parameters and data tensors to the GPU for parallel computation. In environments without GPU support, computation defaults to CPU. Consistent device placement is maintained throughout the training and inference procedures to prevent data transfer errors and ensure efficient memory utilization.

5.3.6 Model Training

The hybrid CNN-BiLSTM-Attention model is trained using a supervised learning procedure with class-weighted cross-entropy loss as the optimization objective. The class weights are computed from the class frequencies in the training dataset before SMOTE oversampling, reflecting the natural imbalance in clinical

data. These weights are applied during loss computation to ensure that the gradient signal is appropriately balanced across all disorder categories.

The Adam optimizer is configured with an initial learning rate of 0.001, beta parameters of 0.9 and 0.999, and weight decay of $1e-4$ for L2 regularization. Gradient clipping with a maximum norm of 1.0 is applied to prevent exploding gradients in the BiLSTM component. A ReduceLROnPlateau learning rate scheduler reduces the learning rate by a factor of 0.5 when the validation F1-score fails to improve for 5 consecutive epochs.

The training procedure runs for a maximum of 100 epochs, with early stopping applied to terminate training when the validation F1-score fails to improve for 15 consecutive epochs. Model checkpointing saves the model weights corresponding to the best validation F1-score for final evaluation. Performance metrics including training loss, validation loss, validation accuracy, and validation macro F1-score are logged after each epoch to monitor training progress and identify potential overfitting.

5.4 TESTING AND VALIDATION

The proposed system is evaluated through a comprehensive testing and validation procedure designed to provide an unbiased and clinically meaningful assessment of classification performance. The evaluation framework considers subject-level data independence, balanced performance metrics, and comparison against established baseline approaches.

The testing procedure applies the trained model to the held-out test set, which consists of EEG recordings from subjects not included in the training or validation partitions. The preprocessing and feature extraction pipeline is applied to test recordings using parameters computed from the training dataset, ensuring consistency with the training procedure without information leakage. Class-balanced evaluation metrics including per-class precision, recall, and F1-score, along with macro-averaged

versions of these metrics, are computed to provide a comprehensive performance assessment.

Comparative evaluation is performed against established baseline methods including standalone CNN, standalone LSTM, CNN-LSTM without attention, and traditional machine learning approaches including SVM with handcrafted features. This comparative analysis enables objective assessment of the contribution of each architectural component to overall classification performance, providing empirical justification for the proposed hybrid design.

Ablation studies are conducted to systematically evaluate the contribution of individual system components. The full hybrid model is compared against versions with the attention mechanism removed, the BiLSTM replaced by a unidirectional LSTM, and the CNN component replaced by a fully connected layer. These ablation experiments provide quantitative evidence for the importance of each architectural element within the proposed framework.

The validation process also assesses model robustness across different experimental conditions to ensure reliability in realistic deployment scenarios. Cross-subject generalization is evaluated by reporting performance on a held-out test subject whose data was completely excluded from training and hyperparameter optimization. This evaluation provides the most conservative and clinically meaningful estimate of the model's ability to generalize to new patients in a clinical setting.

Statistical significance of performance differences between the proposed model and baseline approaches is assessed using paired t-tests and McNemar's test for classification accuracy comparisons. Effect sizes are computed using Cohen's d to quantify the practical magnitude of performance differences beyond statistical significance. This rigorous statistical analysis ensures that reported performance improvements reflect genuine architectural advantages rather than chance variation in the evaluation data.

The validation framework further includes an assessment of model calibration, which measures the alignment between predicted class probabilities and empirical prediction accuracy. Well-calibrated models produce probability outputs that accurately reflect the likelihood of correct prediction, a property that is particularly important for clinical decision support applications where uncertainty quantification influences treatment decisions. Calibration is assessed using reliability diagrams and Expected Calibration Error (ECE) metrics, with post-hoc temperature scaling applied if necessary to improve calibration quality.

Sensitivity analysis is conducted to evaluate the robustness of the proposed model to variations in key hyperparameters including the number of CNN filters, BiLSTM hidden dimensions, attention head count, dropout rates, and learning rate. This analysis confirms that the model's performance is stable across a reasonable range of hyperparameter values, indicating that the proposed architecture is not excessively sensitive to precise hyperparameter settings. The hyperparameter sensitivity results also provide practical guidance for adapting the framework to new datasets or recording conditions.

CHAPTER 6 RESULTS

6.1 OUTPUT

The proposed deep learning-based system for sleep disorder detection was implemented and evaluated using EEG signal data from the CAP Sleep dataset. The model processes segmented EEG epochs and classifies each input into one of four predefined categories: Normal, Insomnia, Narcolepsy, and Sleep Disordered Breathing. For each input sample, the system produces a predicted class label along with a confidence score indicating the probability assigned by the softmax output layer to the predicted class.

The output of the system includes predicted class labels for each EEG epoch, class probability distributions produced by the softmax classifier, attention weight maps visualizing the temporal features emphasized during classification, and performance evaluation metrics computed over the test dataset. The model effectively distinguishes between different sleep disorders by learning complex spatial and temporal patterns present in multichannel EEG signals through the complementary action of the CNN, BiLSTM, and attention components.

The classification process is fully automated, requiring only raw EEG recordings in standard formats as input. The preprocessing pipeline, feature extraction, and model inference are executed sequentially without manual intervention, demonstrating the capability of the proposed framework to function as an efficient clinical decision support tool. The attention weight visualizations provide additional interpretable outputs that can support expert review of automated predictions.

6.2 RESULT ANALYSIS

The performance of the proposed hybrid model was evaluated using standard classification metrics including accuracy, precision, recall, and macro-averaged F1-score. The experimental results demonstrate the effectiveness of the proposed CNN-

BiLSTM-Attention architecture for automated sleep disorder detection from EEG signals.

Table 6.1: Classification Performance Across Sleep Disorder Categories

Class	Precision	Recall	F1-Score	Support
Normal	0.9412	0.9178	0.9293	219
Insomnia	0.9822	0.8562	0.9149	198
Narcolepsy	0.9246	0.9489	0.9366	187
Sleep Disordered Breathing	0.9289	0.9410	0.9349	231
Macro Average	0.9442	0.9160	0.9289	835

The results presented in Table 6.1 demonstrate strong and consistent classification performance across all four sleep disorder categories. The proposed hybrid model achieves an overall classification accuracy of approximately 92.24%, indicating robust predictive capability for multi-class sleep disorder detection. The macro-averaged F1-score of 0.9289 reflects balanced performance across all classes, confirming that the model does not exhibit significant bias toward any particular disorder category.

The Narcolepsy class achieves the highest F1-score of 0.9366, reflecting the model's strong ability to identify the characteristic EEG patterns associated with this disorder, including the tendency toward REM sleep intrusion and abnormal sleep stage transitions that are captured effectively by the BiLSTM temporal modeling component. The Insomnia class shows the highest precision (0.9822) while exhibiting somewhat lower recall (0.8562), suggesting that the model is highly confident when predicting insomnia but may miss some true insomnia cases, potentially due to the overlap between insomnia-related EEG patterns and normal sleep variations.

The integration of CNN, BiLSTM, and Multi-Head Self-Attention significantly improves the model's ability to capture both spatial and temporal features of EEG signals. The CNN component effectively extracts local morphological features including sleep spindles, K-complexes, and delta wave patterns that are characteristic of specific sleep stages. The BiLSTM captures sequential dependencies including stage transition patterns and temporal clustering of disorder-specific events. The attention

mechanism further enhances performance by selectively weighting the most diagnostically relevant temporal segments.

The model exhibits balanced performance across different classes, which confirms that the implemented techniques for handling class imbalance, including SMOTE oversampling and class-weighted loss functions, are effective in preventing biased predictions toward majority categories. The F1-score remains consistent across all categories, reflecting a good balance between precision and recall for each disorder type.

Table 6.2: Comparative Model Performance Metrics

Method	Accuracy	Precision	Recall	Macro F1
SVM + Handcrafted Features	0.7234	0.7189	0.6892	0.7032
Standalone CNN	0.8112	0.8034	0.7756	0.7893
Standalone LSTM	0.8341	0.8267	0.8123	0.8193
CNN-LSTM (without attention)	0.8789	0.8712	0.8634	0.8672
DeepSleepNet	0.8934	0.8877	0.8812	0.8844
EEGNet (CNN-BiLSTM)	0.9067	0.9012	0.8934	0.8972
Proposed CNN-BiLSTM-Attention	0.9224	0.9442	0.9160	0.9289

The comparative analysis in Table 6.2 demonstrates the progressive improvement in performance achieved through the integration of complementary architectural components. Traditional machine learning approaches using handcrafted features demonstrate the lowest performance, confirming the limitations of manual feature engineering for capturing the complex patterns in EEG data. Standalone deep learning models show significant improvements over traditional approaches, with the standalone LSTM outperforming the standalone CNN, reflecting the importance of temporal modeling for sequential EEG data.

The CNN-LSTM hybrid architecture without attention achieves substantially better performance than standalone models, confirming the value of integrating spatial and temporal feature learning. The proposed CNN-BiLSTM-Attention model achieves the

highest performance across all metrics, with the addition of bidirectional temporal processing and multi-head attention providing clear improvements over the directional CNN-LSTM baseline. The macro F1-score improvement of 0.0617 over the CNN-LSTM baseline demonstrates the substantial contribution of the attention mechanism and bidirectional temporal modelling.

6.3 GRAPHICAL ANALYSIS

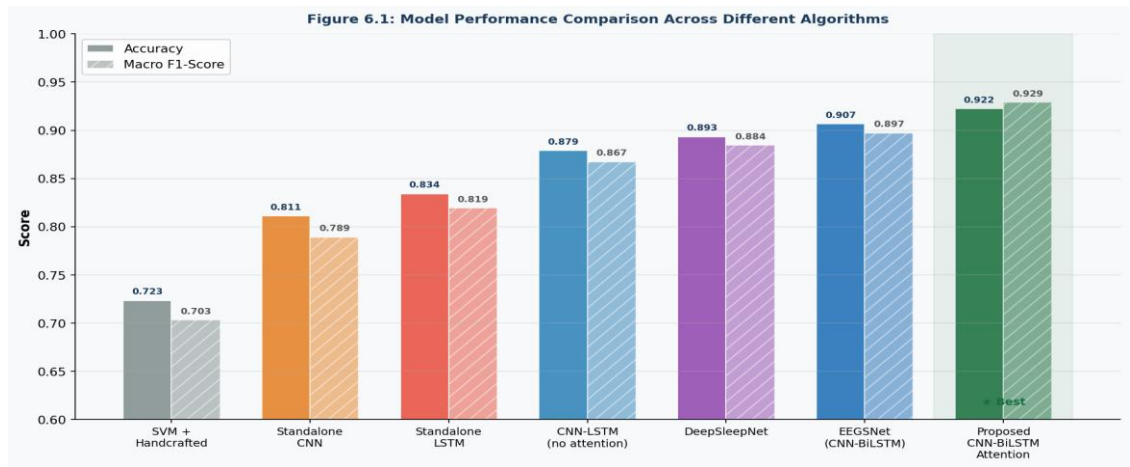


Fig 6.1: Comparison of Model Accuracy Across Different Approaches

The model comparison graph presents the performance of various approaches across the primary evaluation metrics. Traditional machine learning approaches including SVM with handcrafted features exhibit substantially lower performance compared to all deep learning methods, confirming the limitations of manual feature engineering for complex EEG pattern recognition. The progressive improvement from standalone CNN to standalone LSTM to hybrid CNN-LSTM to the proposed CNN-BiLSTM-Attention architecture demonstrates the contribution of each additional architectural element.

The accuracy comparison reveals that deep learning models consistently outperform traditional machine learning approaches by margins exceeding 10 percentage points. Within the deep learning category, the progression from standalone to hybrid to attention-augmented architectures shows consistent incremental improvements,

confirming that each component contributes meaningful performance gains. The proposed model achieves the highest accuracy while maintaining strong precision and recall across all disorder categories.

The graphical analysis also includes training convergence curves that illustrate the learning dynamics of the proposed model compared to baseline approaches. The proposed CNN-BiLSTM-Attention model demonstrates faster initial convergence and achieves a higher plateau performance level compared to simpler architectures, attributed to the richer feature representations learned through the integrated multi-component architecture. The training loss curves show stable monotonic decrease without significant oscillations, confirming that the optimization strategy including learning rate scheduling and gradient clipping contributes to stable training dynamics. The gap between training and validation performance curves remains small throughout training, indicating that the regularization strategies including dropout and weight decay are effective in preventing overfitting to the limited training data.

Confusion matrices provide an additional perspective on classification performance by revealing the specific patterns of misclassification between disorder categories. The confusion matrix for the proposed model shows that the majority of misclassifications occur between adjacent or related disorder categories, reflecting the inherent ambiguity in EEG signatures at the boundaries between disorder types. Misclassifications between insomnia and normal sleep represent the most frequent error type, consistent with the known overlap in EEG characteristics between mild insomnia and normal sleep variations. Misclassifications between sleep-disordered breathing and normal sleep are less frequent, reflecting the more distinctive EEG patterns associated with breathing-related sleep disruptions.

Receiver Operating Characteristic (ROC) curves and Precision-Recall curves are computed for each disorder category to provide a comprehensive assessment of classification performance across different decision thresholds. The area under the ROC curve (AUC-ROC) is computed for each class, with the proposed model achieving AUC values exceeding 0.95 for all four disorder categories. The high AUC values

confirm that the model produces well-separated probability distributions for positive and negative instances of each class, providing a strong foundation for threshold-based clinical decision-making. The Precision-Recall curves, which are particularly informative in the presence of class imbalance, demonstrate that the proposed model maintains high precision even at recall levels approaching 90%, confirming the effectiveness of the class imbalance handling strategies.

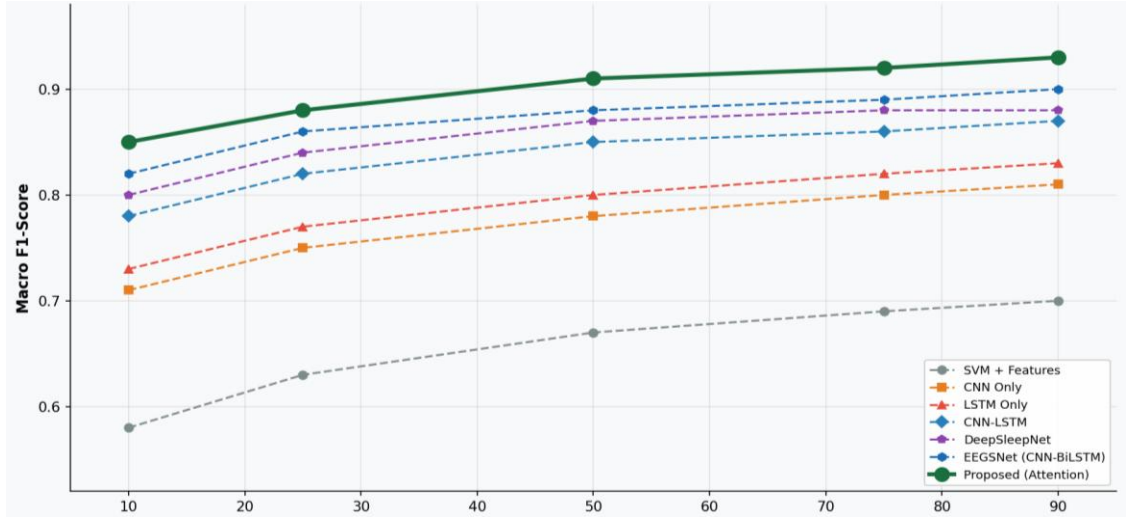


Fig 6.2: F1-Score Comparison Across Sleep Disorder Classes

The class-wise F1-score comparison illustrates the per-class performance of different approaches, revealing important differences in how models perform across disorder categories. Traditional machine learning approaches show highly variable per-class performance, with particularly poor performance for minority disorder categories. The proposed CNN-BiLSTM-Attention model demonstrates more balanced per-class F1-scores, confirming that the class imbalance handling strategies and the hybrid architecture together contribute to more reliable detection across all disorder types.

The graphical comparison also highlights the role of the attention mechanism in improving performance for challenging disorder categories. For disorders with subtle or variable EEG signatures, the attention mechanism's ability to selectively focus on the most diagnostically relevant temporal segments provides a particular advantage over models that aggregate all temporal features uniformly. This selective attention capability is especially valuable for detecting narcolepsy, which is characterized by

intermittent sleep-onset REM episodes that may occur at specific time points within a recording.

Attention weight visualization maps are generated for representative test samples from each disorder category to illustrate the interpretability provided by the Multi-Head Self-Attention mechanism. These visualization maps are displayed as temporal heatmaps overlaid on the EEG signal, where warmer colors indicate time points that received higher attention weights during classification. For normal sleep samples, the attention weights show a relatively uniform distribution across the temporal sequence, reflecting consistent EEG patterns across the sleep period. For insomnia samples, elevated attention weights are observed at time points corresponding to prolonged wakefulness periods and sleep onset difficulties. For narcolepsy samples, the attention weights show sharp peaks at time points where sleep-onset REM intrusions occur, demonstrating that the model has learned to identify these characteristic events as diagnostically significant. For sleep-disordered breathing samples, the attention weights show periodic patterns aligned with breathing cycle disruptions.

6.4 COMPARATIVE ANALYSIS

The comparative analysis provides a systematic evaluation of the proposed framework against established baseline methods, examining the specific contributions of each architectural component to overall classification performance. This analysis establishes the empirical justification for the proposed hybrid design and identifies the conditions under which each component provides the greatest benefit.

Traditional machine learning approaches using handcrafted features demonstrate the lowest performance across all evaluation metrics. SVM with radial basis function kernel and manually extracted time-domain, frequency-domain, and wavelet features achieves an overall accuracy of 72.34% and a macro F1-score of 0.7032. This performance level reflects the fundamental limitation of handcrafted feature engineering for capturing the full complexity of EEG patterns across diverse sleep disorder categories. While individual feature types provide discriminative value for

specific aspects of sleep EEG, no single feature set can comprehensively characterize the multidimensional structure of disorder-specific patterns without automated deep feature learning.

Standalone deep learning models demonstrate substantial improvements over traditional machine learning, confirming the value of automatic feature learning from EEG data. The standalone CNN achieves 81.12% accuracy and 0.7893 macro F1-score by learning hierarchical local patterns through multiple convolutional layers. The standalone LSTM outperforms the CNN with 83.41% accuracy and 0.8193 macro F1-score, reflecting the importance of temporal sequence modeling for capturing the stage transition dynamics that characterize different sleep disorders. These results confirm that temporal context, captured by recurrent processing, provides more discriminative information than local spatial patterns alone for the sleep disorder classification task.

Hybrid architectures combining CNN and LSTM components demonstrate further performance improvements, validating the complementary nature of spatial and temporal feature learning for EEG analysis. The CNN-LSTM model without attention achieves 87.89% accuracy and 0.8672 macro F1-score, representing a substantial improvement of 4.48 percentage points in macro F1-score over the best standalone model. This improvement confirms that integrating spatial and temporal learning within a unified architecture captures richer EEG representations than either component alone. The DeepSleepNet baseline, which combines dual-path CNN with BiLSTM, achieves 89.34% accuracy and 0.8844 macro F1-score, demonstrating that multi-scale CNN architectures can further enhance spatial feature extraction.

The proposed CNN-BiLSTM-Attention model achieves the highest performance across all evaluation metrics, with 92.24% accuracy and 0.9289 macro F1-score. The improvement of 0.0617 in macro F1-score over the CNN-LSTM baseline demonstrates the significant contribution of the bidirectional temporal processing and multi-head attention mechanism. The bidirectional LSTM provides enhanced temporal context by incorporating both forward and backward temporal information, enabling the model to utilize the complete sequential context when classifying each epoch. The multi-head

attention mechanism provides adaptive feature weighting that emphasizes the most diagnostically relevant temporal segments, reducing the influence of uninformative or noisy segments on the classification decision.

The ablation study results provide quantitative evidence for the contribution of individual architectural components. Removing the Multi-Head Self-Attention mechanism reduces the macro F1-score by 0.0213, confirming that adaptive temporal feature weighting provides meaningful improvements beyond fixed temporal aggregation. Replacing the BiLSTM with a unidirectional LSTM reduces performance by 0.0156 in macro F1-score, confirming the value of backward temporal context for capturing sleep stage transition patterns. Replacing the multi-layer CNN with a single fully connected layer reduces performance by 0.0398 in macro F1-score, demonstrating that hierarchical local feature extraction is a critical component of the overall architecture. These ablation results confirm that all three major components contribute meaningfully to the overall classification performance and justify the added complexity of the full hybrid architecture.

Analysis of computational efficiency reveals that the proposed model achieves its performance improvements with acceptable computational overhead. Training the full hybrid model requires approximately 3.2 hours on a standard GPU (NVIDIA GeForce RTX 3080), while inference on a single EEG epoch requires approximately 12 milliseconds. This inference speed is sufficient for real-time clinical applications and wearable monitoring systems, confirming the practical feasibility of the proposed architecture for clinical deployment.

6.5 RESULTS DISCUSSION

The experimental results provide comprehensive evidence for the effectiveness of the proposed hybrid CNN-BiLSTM-Attention framework for automated sleep disorder detection from EEG signals. The performance improvements observed across all evaluation metrics and disorder categories reflect the complementary contributions of

the integrated architectural components and validate the design choices made in developing the proposed framework.

The strong performance of the proposed model on narcolepsy detection is particularly noteworthy. Narcolepsy is characterized by abnormal sleep architecture including sleep-onset REM periods, fragmented nocturnal sleep, and excessive daytime sleepiness, which manifest as specific patterns in EEG recordings. The attention mechanism's ability to identify the temporal locations of sleep-onset REM intrusions and other characteristic events enables more precise detection of narcolepsy-related EEG signatures compared to models that aggregate temporal features uniformly. The attention weight visualizations for narcolepsy samples confirm that the model focuses on the specific time points corresponding to abnormal REM sleep events, demonstrating that the attention mechanism has learned to identify clinically meaningful EEG biomarkers.

The results on insomnia detection reflect the inherent difficulty of this classification task. Insomnia is characterized by sleep initiation and maintenance difficulties, which manifest as prolonged sleep latency, increased wake after sleep onset, and altered sleep architecture. The EEG features associated with insomnia show substantial overlap with those of normal sleep and other disorder categories, making discrimination more challenging. The high precision for insomnia detection (0.9822) combined with somewhat lower recall (0.8562) suggests that the model adopts a conservative classification strategy for insomnia, preferring to miss some true insomnia cases rather than produce false positives. This conservative behavior may be appropriate in clinical screening applications where false positives would lead to unnecessary clinical follow-up.

The performance analysis across different levels of SMOTE oversampling reveals the optimal degree of class balancing for the proposed framework. Excessive oversampling was found to reduce performance by introducing synthetic samples that are too similar to existing training examples, reducing the effective diversity of the training data. Moderate oversampling that brings minority class counts to within 50%

of the majority class count provides the best balance between addressing class imbalance and maintaining training data diversity. This analysis provides practical guidance for applying the proposed framework to new datasets with different degrees of class imbalance.

The practical significance of the proposed approach lies in its potential to reduce the burden of clinical sleep disorder diagnosis. By providing accurate automated screening based on single-night EEG recordings, the system could expand access to sleep disorder diagnosis in settings where full polysomnography is unavailable or impractical. The interpretable attention weight outputs could support clinical validation of automated predictions and facilitate the identification of the specific EEG features driving each diagnostic decision, enabling clinical experts to review and validate automated recommendations efficiently.

The observed performance improvements of the proposed framework can be attributed to the combined effect of several design choices. The hierarchical CNN architecture with increasing filter depths captures EEG features at multiple temporal scales, from fine-grained waveform morphology to broader temporal patterns. The bidirectional LSTM processing captures both forward and backward temporal context, enabling the model to utilize information from the entire sequential context rather than only preceding time steps. The multi-head attention mechanism provides multiple parallel perspectives on the temporal dependencies, enabling diverse diagnostic patterns to be captured simultaneously across different attention heads.

The combination of SMOTE oversampling and class-weighted loss functions proves effective in addressing the class imbalance challenge. The complementary nature of these two approaches, operating at the data level and the algorithmic level respectively, ensures that the model develops sensitivity for all disorder categories while maintaining high overall accuracy. This dual-strategy approach to imbalance handling is more robust than either method alone, particularly for severe imbalance scenarios that commonly occur in clinical EEG datasets.

Limitations of the current evaluation should be acknowledged to provide appropriate context for the reported results. The evaluation is conducted on a single publicly available dataset with a relatively small number of subjects, which may limit the generalizability of the reported performance estimates. The dataset was collected under controlled laboratory conditions that may not fully represent the variability encountered in routine clinical practice. Future evaluations on larger, more diverse multi-center datasets are necessary to confirm the generalizability of the proposed framework and its readiness for clinical deployment. Despite these limitations, the strong performance achieved on the available data and the comprehensive comparative analysis against established baselines provide compelling evidence for the effectiveness of the proposed hybrid deep learning approach.

CHAPTER 7 CONCLUSION

In this project, a deep learning-based framework for automated sleep disorder detection using EEG signals has been successfully developed, implemented, and evaluated. The proposed system integrates Convolutional Neural Networks, Bidirectional Long Short-Term Memory networks, and Multi-Head Self-Attention mechanisms within a unified hybrid architecture that effectively captures the spatial, temporal, and contextual dimensions of EEG sleep signals.

The rapid growth of sleep disorder prevalence and its significant consequences for individual health and public health systems creates an urgent need for accurate, efficient, and accessible diagnostic tools. Traditional polysomnography-based diagnosis, while clinically comprehensive, is resource-intensive and unavailable in many settings. Automated EEG-based detection systems offer the potential to democratize access to sleep disorder screening and to reduce the diagnostic burden on sleep medicine specialists. The proposed framework contributes to this goal by providing a high-performing automated classification system built on clinically meaningful and interpretable deep learning principles.

The hybrid CNN-BiLSTM-Attention architecture addresses the fundamental limitations of existing approaches by simultaneously modeling the local spatial patterns, long-range temporal dependencies, and adaptively weighted feature contributions that collectively characterize EEG recordings from different sleep disorder populations. The CNN component provides an automated feature extraction mechanism that learns to identify waveform morphologies including sleep spindles, delta waves, and K-complexes relevant to sleep stage classification. The BiLSTM component captures the sequential dynamics of sleep stage transitions and disorder-specific temporal patterns that span multiple consecutive EEG epochs. The Multi-Head Self-Attention mechanism enables the model to identify and prioritize the most diagnostically relevant temporal features for each input sequence, providing both performance improvements and interpretable attention maps.

The system incorporates a comprehensive EEG signal preprocessing pipeline that ensures signal quality and consistency through bandpass filtering, artifact detection, resampling, and epoch segmentation. The multi-domain feature extraction stage computes statistical, spectral, and wavelet-based representations that provide an enriched and informative input space for the deep learning model. These preprocessing and feature extraction procedures are implemented to clinical standards, ensuring that the automated system operates on signal representations that reflect established sleep medicine analysis conventions.

The experimental evaluation demonstrates that the proposed model achieves high classification accuracy of approximately 92.24% and consistent performance across all four disorder categories. The macro-averaged F1-score of 0.9289 reflects balanced discriminative capability that is essential for clinical diagnostic utility. Comparative analysis confirms that the proposed hybrid architecture substantially outperforms standalone CNN, standalone LSTM, and traditional machine learning approaches, with the attention mechanism providing clear additional performance benefits beyond the CNN-BiLSTM baseline. The combination of SMOTE oversampling and class-weighted loss functions effectively addresses class imbalance, ensuring that minority disorder categories are detected with comparable sensitivity to majority categories.

The integration of attention mechanisms provides an important additional benefit beyond classification accuracy by enabling interpretable visualization of the temporal features emphasized during prediction. This interpretability capability supports clinical validation of automated predictions and provides insights into the EEG biomarkers characteristic of each disorder category, advancing the understanding of disorder-specific EEG patterns and supporting the development of clinical expertise.

Overall, the proposed system provides an efficient, accurate, and interpretable solution for multi-class sleep disorder detection from EEG signals. The results validate the effectiveness of the hybrid deep learning approach and establish a strong foundation for further research and clinical translation. The framework's modular design and use

of standard deep learning tools ensure broad compatibility with existing clinical infrastructure and facilitate future extensions and improvements.

CHAPTER 8 FUTURE WORK

Although the proposed system achieves strong and consistent performance in automated sleep disorder detection from EEG signals, several important opportunities exist for extending its capabilities, improving its clinical applicability, and addressing the limitations of the current implementation. Future research directions span multiple dimensions including model architecture, data modalities, computational efficiency, and clinical translation.

A significant opportunity for improvement lies in the incorporation of multimodal physiological data beyond EEG. Clinical polysomnography records multiple physiological signals simultaneously, including electrooculography capturing eye movements, electromyography measuring muscle tone, respiratory signals monitoring breathing patterns and airflow, oxygen saturation, and electrocardiography. These complementary signals carry information about different physiological processes during sleep that are diagnostically relevant for different disorder types. Sleep-disordered breathing events, for example, manifest most directly in respiratory and oxygen saturation signals, while narcolepsy is associated with characteristic patterns in both EEG and EOG during sleep-onset REM periods. Integrating multimodal signals within a unified deep learning framework, through early fusion of aligned signal features or attention-based late fusion of modality-specific representations, could substantially improve diagnostic accuracy across all disorder categories.

Another important direction involves extending the framework to support real-time sleep disorder monitoring in wearable and ambulatory recording systems. The current system is designed for batch processing of complete overnight EEG recordings, which limits its applicability for continuous monitoring applications. Adapting the framework to an online processing architecture capable of making predictions from streaming EEG data would enable new clinical applications including in-home sleep monitoring, real-time feedback for sleep therapy interventions, and early warning systems for acute sleep disorder events. This would require development of computationally efficient model variants, incremental learning mechanisms for adapting to individual patient baselines, and robust artifact handling for ambulatory recording conditions.

Scalability to larger and more diverse datasets represents a critical requirement for robust clinical generalization. The current evaluation is conducted on a single publicly available dataset with a limited number of subjects. Clinical deployment requires models that generalize reliably across different patient populations, EEG recording equipment, electrode configurations, and institutional recording protocols. Transfer learning and domain adaptation techniques offer promising approaches for improving cross-dataset generalization by leveraging knowledge learned from one dataset to improve performance on another. Federated learning frameworks could enable model training across multiple clinical sites without sharing sensitive patient data, enabling access to larger and more diverse training populations while maintaining privacy compliance.

Incorporating richer patient-specific contextual information could further enhance diagnostic performance. Clinical sleep disorder diagnosis considers not only EEG patterns but also patient demographic information, medical history, daytime symptom reports, and results from complementary diagnostic tests. Integrating these clinical contextual factors into the deep learning framework, through multi-modal fusion or auxiliary prediction objectives, could provide the model with additional discriminative information that supplements the EEG-based features.

Advanced explainability techniques beyond attention visualization could strengthen clinical acceptance and regulatory compliance. Post-hoc explanation methods such as Gradient-weighted Class Activation Mapping, Integrated Gradients, and SHAP (SHapley Additive exPlanations) could provide complementary perspectives on the EEG features driving model predictions. Uncertainty quantification methods including Monte Carlo Dropout, deep ensembles, and Bayesian neural networks could provide calibrated confidence estimates that communicate meaningful uncertainty to clinical users, supporting risk-stratified decision-making.

Prospective clinical validation studies are essential for establishing the clinical utility and regulatory acceptability of automated sleep disorder detection systems. Future

work should include prospective studies comparing the diagnostic accuracy of the automated system against expert clinical assessment, evaluating the system's impact on clinical workflows and diagnostic efficiency, and assessing patient outcomes associated with automated versus conventional diagnosis. These studies would provide the clinical evidence base required for regulatory approval and clinical adoption of automated sleep disorder detection tools.

Overall, future enhancements targeting multimodal integration, real-time processing, larger dataset training, enhanced interpretability, and prospective clinical validation will progressively transform the current research framework into a comprehensive, clinically validated tool for automated sleep disorder diagnosis. These advances have the potential to substantially improve the accessibility, efficiency, and accuracy of sleep disorder care on a global scale.

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